



EE 613: Wireless Communication

Course Code	EE 613
Course Name	Wireless Communication
L-T-P-C	3-0-0-3
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Cellular Communications:** Introduction to Cellular Communications, Frequency reuse, Channel assignment strategies, Hand-off strategies, Interference and system capacity, Trunking and Grade of services, Improving Coverage and capacity in cellular systems (cell splitting, sectoring, microcell Zone, Tele-traffic Theory)
- **Large scale path loss, small scale fading and Diversity:** Wireless Channel Modeling, Path loss, Hata, Okumura Models, Shadowing, Diffraction Knife Edge models, Fast Fading, Rayleigh/Ricean Fading Channels, BER Performance, Radio Power budgeting, Diversity, BER Performance with diversity, Types of Diversity, RMS Delay Spread, Doppler Fading, Jakes Model, Autocorrelation, Jakes Spectrum, Impact of Doppler Fading
- **Modems for Wireless Communications:** Analog modulation, Digital Modulation, Pulse Trains and Pulse shaping, ASK, BPSK, M-ary Modulation, Constant Envelope Modulation Techniques, M-ary phase modulation, M-ary QAM, O- QPSK, Pi/4 QPSK, MSK, GMSK, Shannon theorem, Channel capacity in Rayleigh fading
- **CDMA:** Introduction to CDMA, PN Sequences, DS CDMA, FH CDMA, Multipath diversity, RAKE Receiver, CDMA Receiver Synchronization
- **WirelessStandards:** AMPS, IS 54, GSM, CDMA 2000